

Autonomous Theory Construction Laboratory

Lecture 6: Auxiliary Terms and Local Gates

Garam Yun

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Overview

Local gates restrict auxiliary terms to the region of data space in which they are supported.

1 Problem definition

The scientific question and the admissible output are specified before any search begins.

$$r_a \rightarrow cluster(x_a, E_a, \rho_a, I_a)$$

2 Data and model interface

Data schemas and model interfaces are treated as part of the method, not as post-processing details.

$$\Delta T_a = G_c(x_a) f_a(\eta_c)$$

3 Search and evaluation

Proposal and evaluation are kept separate so that a candidate cannot alter its own test.

$$|\partial \Delta T_a / \partial \eta_c| < \epsilon_{outsidec}$$

4 Reproducibility

Environment, data version, random seed, and decision record are sufficient to reproduce the reported run.

5 Exercise

The exercise asks for a small implementation and an error analysis rather than a polished narrative.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Yun, ch. 6
- Gate Library GL-26
- Auxiliary Term Registry